Scale-Aware Conditional Privacy Audits for Graph Neural Network Prediction APIs

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***Abstract*—We study *conditional* membership-privacy audits for graph ML prediction APIs: leakage depends on structure, architecture, and split—not on "GNNs" or node count alone. SPAB is a scale-aware audit protocol and release artifact reporting Acc, LiRA (primary), TPR@1%FPR, train time, and memory as $(h, \rho, M, n) \mapsto$ metrics, with NeighborLoader systems numbers through ogbn-arxiv ($\sim$169k). *SPAB's Volume contribution is systems cost plus trustworthy negative controls, not proof that large graphs always leak*: under the official OGB split, ogbn-arxiv score MIAs sit near chance. Privacy findings concentrate where risk is predicted under one locked SAMI config (λ=0.5, σ=0.35): Cora GraphSAGE (LiRA $0.567 \rightarrow 0.510$), Citeseer ($0.596 \rightarrow 0.537$), and Chameleon ($0.623 \rightarrow 0.551$) at matched Acc. Near-chance Volume/multi-k negatives (ogbn-arxiv with n_{shadows}=4, PubMed, Computers) are the Big Data finding—scale alone does not create score MIA here. Actor and a GCN hard cell are boundaries; Acc–LiRA Pareto plots vs. GTD/LBP/MaskArmor make the joint frontier explicit.**

***Index Terms*—graph neural networks, membership inference, privacy audit, large-scale graph analytics, ogbn-arxiv, LiRA**

I. INTRODUCTION

A. Motivation

Modern data platforms serve GNN node classifiers [8], [9] behind prediction APIs over citation, payment, and referral graphs. Those APIs create a membership-privacy surface [1]–[4]. For large-scale deployments the right question is not "do GNNs leak?" but *under which structure, architecture, and scale does leakage appear, at what systems cost, and where should a finite privacy budget be spent?*

a) What we release.: **SPAB** is a scale-aware *audit protocol* and release CSV (not a claim of community benchmark adoption): LiRA-primary rows with systems fields where measured, through ogbn-arxiv. **SAMI** is one locked-config empirical hardening (LTE heuristic + ϕ-alignment). A synthetic LOO plot is supporting evidence only [5].

b) Empirical spine.: Fig. 3 separates *audit regimes* (not a single n-axis claim): near-chance under official ogbn-arxiv, PubMed, and Computers; elevated LiRA where we intervene (Cora, Citeseer, Chameleon) plus Actor as a heterophilic boundary; a synthetic cell is a controlled stress test only. **Volume ownership:** the Big Data result is that score MIA stays near chance at 169k (ogbn, n_{shadows}=4) and on Computers (13.7k)—*null at scale*—while NeighborLoader systems cost is measurable; this does *not* claim large graphs always leak. The privacy story is locked-config SAMI on Cora, Citeseer, and Chameleon (LiRA at matched Acc vs. none/GTD/MaskArmor); Fig. 4 places these against LBP's Acc–LiRA tradeoff (LBP can win raw LiRA at large Acc cost). Actor and a GCN hard cell remain boundaries.

B. Gap and related defenses

LBP adds global output noise [5]; GTD uses alternate training and loss flattening [18]; MaskArmor truncates confidence tails [19]; GPS targets *attribute* leakage via structure [32]; GAP/NAP-GNN provide ε-DP [12], [33]. What is incremental here is not "structure matters," but a *scalable conditional audit* with systems metrics plus a *risk-budgeted* empirical hardening that allocates effort where local topology predicts membership signal—without claiming DP.

C. Contributions

1) **SPAB (primary).** A scale-aware *audit protocol* and release artifact (not a community benchmark claim): fixed columns, NeighborLoader systems through ogbn-arxiv as negative control, regime map with explicit split tags.
2) **One defense (SAMI).** One locked config (λ=0.5, σ=0.35) for citation/heterophilic cells; LiRA wins on Cora, Citeseer, and Chameleon at matched Acc; Acc–LiRA Pareto vs. LBP/GTD; Actor boundary; $-$LTE Acc collapse; honest GCN hard cell.
3) **Supporting evidence (not a law).** Held-out synthetic LOO plot interpreting where score MIAs rise—one figure, demoted from a formal claim.

The remainder of the paper proceeds as follows. Section II reviews related work; Section III the threat model. Section IV states the supporting predictive regularity; Section V presents SAMI. Sections VI–VII evaluate; Section VIII notes boundaries.

II. BACKGROUND AND RELATED WORK

A. Membership inference on classical models

Shokri et al. [1] introduced shadow-model MIAs; Salem et al. [2] showed that simple confidence/entropy features often suffice. Yeom et al. [3] linked attack advantage to overfitting. Carlini et al. [4] argued for likelihood-ratio attacks (LiRA) and low false-positive operating points as security-relevant metrics. Related extraction and inversion attacks [22], [23], [30] further motivate treating prediction APIs as privacy surfaces. We adopt both confidence-based ϕ-attacks (continuity with prior GNN work) and Gaussian LiRA as the primary privacy metric, and we report calibration error because overconfident probabilities are a recurring membership cue [15], [31].

B. Membership inference on graphs

Olatunji et al. [5] demonstrated node-level black-box MIAs on GNNs and proposed LBP and neighborhood sampling defenses. He et al. [35] systematically study node-level MIAs under varying topology knowledge. Subsequent work studies graph-level membership [6], embedding leakage [21], and broader attack taxonomies [7]. GTD [18] targets the transductive setting with alternate training and loss flattening. Khosla [36] shows that training-graph construction and inference-time edge access directly modulate membership advantage—orthogonal to our score-API audit protocol. GPS [32] studies structure→*attribute* leakage; we target score-based *membership*. Our threat model matches the score-based black-box node setting; we do not claim white-box, structural-modification, or federated attack coverage.

C. Architectures, regularization, and privacy

GCNs [8] apply spectral neighborhood averaging; GraphSAGE [9] uses mean aggregation and is often more robust under heterophily. DropEdge [10] randomly deletes edges during training. DP-SGD and DP-GNNs [12], [13] provide formal privacy; SAMI is an empirical mitigation and does **not** claim differential privacy. Homophily modulates the train/test ϕ-gap [16], [17]: under low h and sparsity, supervised memorization widens membership cues—motivating risk-weighted budget allocation.

D. Positioning

Primary claim: **SPAB**—conditional privacy+systems audits for graph ML APIs at scale. SAMI is one optional hardening—not three equal breakthroughs. Table I situates the gap: we do not beat GAP on guarantees; we report deployable LiRA-primary audits with systems cost through ogbn-arxiv and a locked risk-weighted defense that wins LiRA at matched Acc on Cora, Citeseer, and Chameleon. We follow modern MIA guidance [4], [34]: defense-aware shadows, TPR@low FPR, label-only gap attacks, and a tuned DP-SGD Pareto anchor.

III. Problem Setup

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n\times d}$, and labels $y \in \{0, \ldots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{S}$ with ratios $40\%/20\%/40\%$ (resampled per seed). Training is *transductive*: every node participates in message passing; only $\mathcal{T}$ enters the cross-entropy loss. Table III summarizes notation.

TABLE I
PRIOR-ART CONTRAST (SETTING / STRUCTURE / DP / SCALE). GAP COLUMN IS OUR CONTRIBUTION RELATIVE TO THAT LINE.

Work	Setting	Struct.	DP?	Scale	Our gap
Olatunji+LBP [5]	Score MIA	No	No	Citation	Systems+LiRA protocol
He et al. [35]	Node MIA	Topo.	No	Citation	Audit recipe+defense
GTD [18]	Transductive	No	No	Citation	Risk-weighted alt.
GPS [32]	Attr. leak	Yes	No	Med.	Membership scores
Khosla [36]	MI risk	Yes	No	Med.	API audit+cost
GAP [12]	DP GNN	Partial	Yes	Large	Emp. matched Acc

TABLE II
METHOD CONTRAST (GUARANTEE TYPE / UTILITY / STRUCTURE / MIA SCORE COVERAGE).

Method	Guarantee	Utility	Structure	MIA scores
SAMI	Empirical	Low–mod.	Yes (LTE)	Score black-box
GAP	ε-DP (agg.)	Often high	Partial	Broad via DP
NAP-GNN	ε-DP (imp.)	Tuned	Yes (imp.)	Broad via DP
LBP	Empirical	High cost	No	Posterior MIA
GTD	Empirical	Low–mod.	No (global)	Loss-gap MIA

TABLE III
PRIMARY NOTATION.

Symbol	Meaning
n, d, C	Nodes, feature dimension, classes
X, y	Features and labels
$\mathcal{T}, \mathcal{V}, \mathcal{S}$	Train / val / test node sets
$\hat{p}_v$, z_v	Softmax posterior and logits at v
$\phi(v)$	Four-dimensional attack feature map
r_v	LTE leakage risk in $[0, 1]$
$h(G)$, ρ	Observed homophily, edge density
ECE	Expected calibration error (B=10)

B. Threat model

Definition 1 (Black-box node membership adversary)**.** *The adversary queries a deployed node classifier and observes softmax vectors $\hat{p}_v$ (and, for evaluation, true labels y_v used to build score features, following standard MIA methodology [2], [5]). The adversary's goal is to decide whether $v \in \mathcal{T}$ or $v \in \mathcal{S}$. Validation nodes are excluded from the final membership evaluation to avoid contamination with early-stopping / defense proxies.*

Equivalently, the adversary plays a membership game: a challenger samples v uniformly from $\mathcal{T} \cup \mathcal{S}$, reveals $(\hat{p}_v, y_v)$, and the adversary outputs a bit $\hat{m} \in \{0, 1\}$. Advantage is $\Pr[\hat{m} = \mathbb{1}[v \in \mathcal{T}]] - \frac{1}{2}$, estimated in aggregate by AUROC over many queries. We emphasize AUROC (and TPR at low FPR) rather than balanced accuracy alone, following Carlini et al. [4].

The adversary has no access to gradients, hidden embeddings, or training randomness beyond what is revealed by posteriors. White-box, link-stealing, and structure-manipulation attacks are out of scope [7].

a) Threat-model boundary (partial knowledge).: An adversary who knows LTE is used could try to reverse-engineer high-risk neighborhoods from public topology and concentrate queries there; an adversary with black-box access to known non-members can calibrate ϕ thresholds; and because risk-scaled Laplace release is randomized, a multi-query adversary may average K independent releases to reduce noise. We evaluate defense-aware shadows (same training defense and release API), multi-query averaging ($K \in \{1, 5, 20\}$), and a nonlinear MLP-ϕ attacker. We do *not* claim robustness to white-box gradients, adaptive poisoning of the graph, or unbounded query budgets without a session-level noise/query policy.

C. Attack feature map

Following Salem-style score attacks [2], define

$$\phi_1(v) = \max_c \hat{p}_{v,c}, \qquad \phi_2(v) = \hat{p}_{v,y_v}, \tag{1}$$

$$\phi_3(v) = -\sum_{c=1}^{C} \hat{p}_{v,c} \log(\hat{p}_{v,c} + \varepsilon), \quad \phi_4(v) = -(1-\phi_1)\log(\phi_1 + \varepsilon). \tag{2}$$

Intuitively, ϕ_1 and ϕ_2 capture confidence, ϕ_3 entropy, and ϕ_4 a modified-entropy margin. These are exactly the features our defense aligns, so the training objective matches the attack surface.

D. Metrics

Utility (higher better): test accuracy, macro-F1, macro AUROC. **Privacy** (lower attack success better): confidence-attack AUROC (logistic regression on ϕ), threshold AUROC on ϕ_2, shadow-model AUROC, and **LiRA AUROC** with TPR at FPR $\in \{0.001, 0.01\}$ [4]. We also report ECE [15] because overconfidence is a common membership cue. An attack AUROC of 0.5 is chance; values near 1.0 indicate severe leakage.

Remark 1. *Because the setting is transductive, non-members are still present in the graph during training. This is slightly weaker than inductive isolation of entire nodes, but remains the dominant practical semi-supervised GNN deployment mode and still admits nontrivial membership distinguishability through the supervised loss asymmetry.*

IV. Supporting Evidence: Structure Predicts Score MIA

Prior work already links structure to GNN membership risk [5], [32]. We do *not* claim a new law. As supporting evidence for interpreting SPAB cells, a ridge fit on a controlled $(h, \rho, \mathrm{SNR}, M)$ synthetic grid yields leave-one-regime-out Spearman $\approx$0.70 (MAE $\approx$0.038) for confidence-attack AUROC (Fig. 1). Real graphs separate qualitatively: PubMed near chance; Actor high conf/LiRA; Planetoid shows MLP>GNN on conf AUROC (feature channel), which the SNR axis reproduces inside the same design. Coefficients are modest; we treat this as an *interpretive synthetic grid* for audits, not a contribution equal to SPAB.

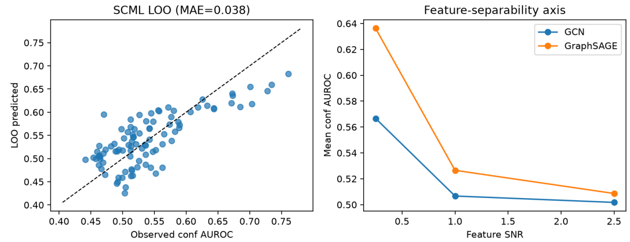

Fig. 1. Supporting LOO plot (not a formal theory): predicted vs. observed conf AUROC on held-out synthetic regimes.

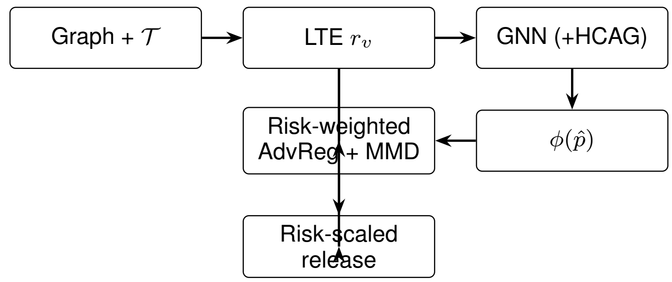

Fig. 2. SAMI: LTE heuristic risk $\rightarrow$ risk-weighted ϕ-alignment; optional release.

V. SAMI: Risk-Weighted ϕ-Alignment (One Defense)

A. Design principle

SAMI estimates per-node structural risk r_v with a three-term heuristic (**LTE**: inverse degree, local heterophily, supervised-neighbor fraction; equal weight, min–max), then applies risk-weighted ϕ-indistinguishability. HCAG and risk-scaled Laplace release are optional. LTE is a *proxy*, not a learned leakage estimator; we report Spearman(r_v, $|\phi_2$-gap) across datasets (Table XVI). Removing LTE (uniform weights) collapses Acc ($\approx$0.11 on Cora) while privacy barely moves—validity for utility, not proof that LTE is the optimal privacy targeting.

B. Module 1: Leakage Topology Estimator (LTE)

For each node v, compute three non-negative signals that use only G, $y|_{\mathcal{T}}$, and the train mask (never test labels):

$$s_v^{\deg} = \frac{1}{1 + \deg(v)}, \tag{3}$$

$$s_v^{\mathrm{het}} = 1 - \hat{h}_v, \qquad \hat{h}_v = \frac{\sum_{u \in \mathcal{N}(v) \cap \mathcal{T}} \mathbb{1}[y_u = y_v]}{\max(1, |\mathcal{N}(v) \cap \mathcal{T}|)}, \tag{4}$$

$$s_v^{\mathrm{sup}} = \frac{|\mathcal{N}(v) \cap \mathcal{T}|}{\max(1, \deg(v))}. \tag{5}$$

Nodes with no train neighbors receive mid-scale heterophily risk (0.5) rather than a pathological maximum. Define the raw score as the mean of (3)–(5) and min–max normalize to $r_v \in [0, 1]$. **Ablation** $-$**LTE** replaces r_v with the constant 1 (uniform weighting).

Algorithm 1 SAMI training and release (single trial)

Require: Graph G, masks $\mathcal{T}, \mathcal{V}, \mathcal{S}$, model f_θ, λ, η, σ, epochs T

Compute LTE risks $\{r_v\}$ via (3)–(5); normalize to $[0, 1]$
for epoch $= 1$ to T **do**
  Forward f_θ (with HCAG gates if enabled)
  Update D on detached ϕ to classify $\mathcal{T}$ vs $\mathcal{V}$
  Update θ by minimizing (6)
end for
Release $\tilde{p}_v \leftarrow \mathrm{renorm}\big(\mathrm{softmax}(z_v/T_v) + \mathrm{Lap}(0, \sigma r_v)\big)$
Evaluate MIAs using members $\mathcal{T}$ and non-members $\mathcal{S}$ only

C. Module 2: Risk-weighted ϕ-indistinguishability

Let D be a small MLP discriminator on $\phi \in \mathbb{R}^4$. Each epoch: (i) update D to separate $\mathcal{T}$ from $\mathcal{V}$ on *detached* ϕ; (ii) update the GNN to minimize

$$\begin{aligned} \mathcal{L} = \mathcal{L}_{\mathrm{CE}}(\mathcal{T}) + \lambda\, \mathbb{E}_{v\sim\mathcal{T}\cup\mathcal{V}}\big[r_v\, \mathrm{BCE}\big(D(\phi(v)), 0\big)\big] \\ + \lambda\, \mathrm{MMD}_r(\phi_{\mathcal{T}}, \phi_{\mathcal{V}}) + \eta\, \mathbb{E}_{v\in\mathcal{T}}\big[r_v\big(-\mathrm{H}(\hat{p}_v)\big)\big], \end{aligned} \tag{6}$$

where the AdvReg term pushes all scored nodes toward the non-member decision of D, MMD_r matches risk-weighted first and second moments of ϕ between train and validation, and the entropy term discourages overconfidence on high-risk supervised nodes. **Critical hygiene:** validation—not test—is the non-member proxy, so the final $\mathcal{T}$ vs. $\mathcal{S}$ evaluation is uncontaminated.

D. Module 3: Homophily-Conditioned Aggregation Gate (HCAG)

For gated GCN/SAGE variants, a per-edge gate

$$g_{uv} = \sigma\big(\mathrm{MLP}([h_u \| h_v \| r_u \| r_v])\big) \tag{7}$$

reweights second-layer aggregation. Gates are initialized near 1 (open) for stable early training. HCAG is the architectural counterpart to LTE: it softens message paths likely to create train/test posterior asymmetry under heterophily.

E. Module 4: Risk-scaled release

At inference we optionally apply temperature $T_v = 1 + \beta r_v$ and add Laplace noise with per-node scale σr_v, then clip and renormalize. Full SAMI uses locked σ=0.35 (Table IX). The identical API transform is applied to shadow models so attackers face the defended interface.

F. Complexity and systems cost

LTE is a single sparse pass over edges: $O(|E|)$. HCAG adds a small MLP evaluation per edge in the gated layer: $O(|E|d_h)$. The discriminator operates on $O(n)$ four-dimensional vectors and is negligible beside GNN training. Risk-scaled release is $O(nC)$ and matches the cost tier of LBP. Shadow/LiRA evaluation multiplies training cost by n_{shadows} and should be reported separately from defense overhead in deployment audits. On PubMed GraphSAGE, wall-clock training rises from $\approx$ 7s (`none`) to $\approx$ 22s (full SAMI with HCAG) on CPU (Table IV). **ogbn-arxiv** ($\sim$169k nodes) is enabled via NeighborLoader (HCAG off at Volume); we report Acc/conf/LiRA, train seconds, peak RSS, and defended API QPS in Section VII and Table XII. LiRA shadow counts are *protocol-sized by wall-clock*, not attack-tuned: confirmatory citation cells use n_{shadows}=4; Volume ogbn uses n_{shadows}=2 from the smoke gate in `results/ogbn_smoke_timing.json` (one-shadow wall-clock projection under a $\sim$12–18h CPU budget). We do not claim a camera-ready "target 8" unless that run is executed.

TABLE IV
SYSTEMS OVERHEAD ON PUBMED GRAPHSAGE (CPU, SEED 42, NO LIRA SHADOWS).

Defense	Wall (s)	Train (s)	Peak MB
none	6.9	6.4	5.9
SAMI (−HCAG)	13.9	13.4	6.2
SAMI (full)	22.1	21.6	6.2

TABLE V
DEFENSE HYPERPARAMETERS (FIXED ACROSS CONFIRMATORY RUNS).

Defense	Setting
DropEdge	$r_{\mathrm{de}} = 0.3$
Label smoothing	$\alpha = 0.1$
Early stopping	patience 15, $\delta = 10^{-4}$
Confidence masking	top-$k = 2$
Edge sparsification	$r_{\mathrm{es}} = 0.2$
LBP	Laplace scale 0.3
GTD-style	$\gamma = 1.0$, stage-1 fraction 0.5, pseudo-conf 0.8 (citation); Volume-sa
SAMI	$\lambda = 0.5$, $\sigma = 0.35$, warmup 5, HCAG on (locked)

G. What SAMI is not

SAMI is not a substitute for DP-SGD when a cryptographic or legal privacy budget is required. It is a structure-aware hardening of the prediction API against score-based membership adversaries, designed to be drop-in compatible with standard GCN/SAGE training loops and YAML-driven experiment grids.

VI. EXPERIMENTAL PROTOCOL

A. Pre-declared evaluation rules

We freeze the protocol before interpreting confirmatory results [4]: primary privacy = LiRA AUROC (+ TPR@FPR); primary utility = test accuracy / macro-F1 / macro AUROC; secondary = confidence/threshold/shadow AUROC and ECE. Defense hyperparameters are **not** selected to minimize *test* attack AUROC.

B. Fair training

All methods share Adam settings, epoch budget, and architecture width unless a baseline's definition requires a specific mechanism (e.g., DropEdge rate 0.3, LBP scale 0.3, GTD flattening weight γ=1). SAMI uses the locked config λ=0.5, warmup 5, σ=0.35 (Table IX). Table V lists defense hyperparameters held fixed across the confirmatory grid.

C. Baselines

`none`; DropEdge; label smoothing (α=0.1); early stopping; confidence masking (top-2); MaskArmor-style top-1 masking; edge sparsification; **LBP**; **GTD-style** alternate+flattening; structure-blind **AdvReg**; and a DP-SGD / DP-GNN *reference point* on the Pareto plot (formal privacy, typically steep utility cost)—not claimed as a SAMI competitor under the same utility budget.

D. Attacks and statistics

Confidence, threshold, shadow, and Gaussian LiRA with n_{shadows}=4 in the confirmatory citation core. Volume (ogbn-arxiv) uses n_{shadows}=2 because a one-shadow smoke gate projected the 4-defense$\times$3-seed grid outside the CPU budget at n=4; the reduction is a *systems* choice, not a privacy-metric cherry-pick. **Defense-aware shadows:** every shadow applies the same training defense and release transform as the target (including SAMI noise), so confidence/threshold/shadow/LiRA attackers are not evaluated against a mismatched undefended API. We additionally report an MLP-ϕ attacker and multi-query averaging ($K \in \{1, 5, 20\}$) against randomized release. Five seeds $\{42, 123, 456, 789, 1024\}$ on citation/Actor cells; three seeds on Volume; paired t-tests vs. `none`; Holm–Bonferroni adjustment within the confirmatory family; 95% bootstrap CIs over seeds *and* bootstrap CIs on the paired effect ΔAUROC itself; effect sizes ΔAUROC and Δaccuracy.

a) Power analysis.: On Cora GraphSAGE with locked σ=0.35 and n_{shadows}=8 (3 seeds), LiRA ΔAUROC (SAMI$-$none) has mean -0.153 with paired SD 0.016; we treat this as a *sensitivity check* (not a fully powered multi-seed LiRA study) and still report five-seed means elsewhere. Conf ΔAUROC remains secondary.

E. Ablations

Full SAMI; $-$LTE; $-$Adv/MMD (λ=0 with release noise); $-$HCAG; temperature-only; AdvReg.

F. Negative controls

We include (i) regimes where leakage is already near chance (PubMed; high-homophily dense synthetics), where large AUROC drops would be spurious; (ii) comparison against a random-guess floor of 0.5; and (iii) reporting of train–test accuracy gaps alongside attack AUROC to connect overfitting diagnostics [3] with membership risk.

VII. Results

Unless noted, citation numbers are five-seed means from `results/core_results.csv`; Volume/Actor/high-risk cells use the seeds stated in each table. The public SPAB report (`results/spab_report.csv`) matches these rows column-for-column.

A. SPAB at a glance

Table VI summarizes LiRA under the locked SAMI config vs. none and the best LiRA-only baseline (often LBP). Ideal = low LiRA at high Acc; Fig. 4 shows the joint frontier.

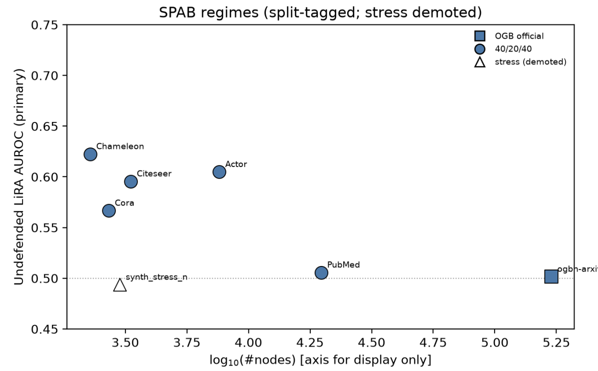

Fig. 3. SPAB regime sketch (undefended *LiRA* AUROC where available, else conf): markers tagged by split protocol. Points are not commensurate on a single causal n-axis—ogbn uses the official OGB split; citation/Actor use 40/20/40; the synthetic stress cell is demoted (open marker).

TABLE VI
SPAB AT A GLANCE (GRAPHSAGE; LOCKED SAMI σ=0.35 UNLESS NOTED). Δ = SAMI$-$NONE LIRA.

Dataset	none	SAMI	Δ	best base.	Tag
Cora	0.567	0.510	-0.057	GTD 0.538	win
Citeseer	0.596	0.537	-0.059	LBP 0.530	win†
Chameleon	0.623	0.551	-0.072	LBP 0.544	win†
Actor	0.605	0.552	-0.053	LBP 0.543	partial
ogbn-arxiv	0.503	0.503	0.000	—	neg. ctrl
Computers	0.499	0.500	$+0.001$	—	neg. ctrl

†LBP lower on LiRA alone at large Acc cost; SAMI is the joint (Acc, LiRA) point vs. none.

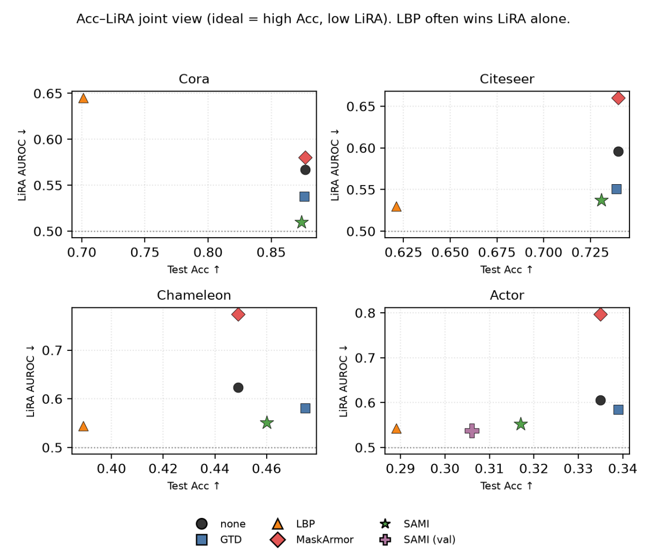

Fig. 4. Acc vs. LiRA on Cora/Citeseer/Chameleon/Actor (none/GTD/LBP/MaskArmor/SAMI). Ideal = bottom-right. LBP often wins LiRA alone; locked SAMI is competitive on the joint frontier.

B. Flagship cell: Cora GraphSAGE (LiRA primary)

Table VII and locked config Table IX. **Primary:** on Cora GraphSAGE, locked SAMI improves LiRA 0.567 $\to$ 0.510 and TPR@1%FPR 0.033 $\to$ 0.003 vs. none, beating GTD

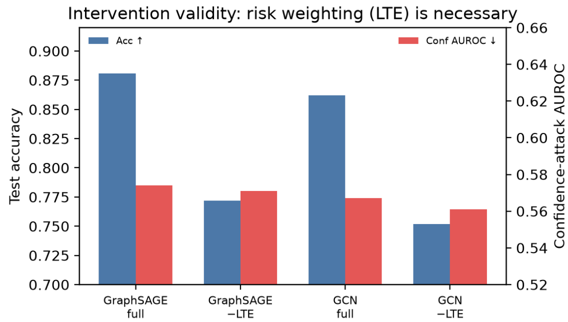

Fig. 5. Full SAMI vs. −LTE on Cora: Acc collapses without risk weighting.

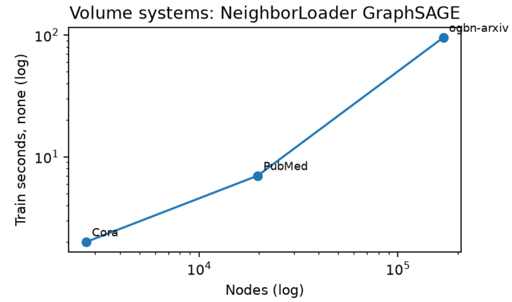

Fig. 6. Train time vs. graph size (ogbn systems).

(LiRA 0.538) and LBP (LiRA 0.645 at Acc 0.701); MaskArmor matches conf but inflates LiRA to 0.580. Conf AUROC is secondary (0.650 → 0.563). **Scope:** locked SAMI also wins LiRA on Citeseer and Chameleon at matched Acc (Tables VI–XV); Fig. 4 shows LBP can still win LiRA alone at large Acc cost. Cora GCN LiRA change remains tiny (0.539 → 0.533). **Hard cell (up front):** `synthetic_low_sparse` GCN fails our success bar (Table VIII).

C. Citeseer: locked config suffices

Under locked $(\lambda, \sigma) = (0.5, 0.35)$, Citeseer GraphSAGE LiRA 0.596 → 0.537 (Δ=−0.059) at Acc 0.740 → 0.731 (within 0.01). A val-Acc sensitivity grid selects $(1.0, 0.25)$ with essentially identical LiRA (0.538) and Acc 0.744—we therefore *lead with the locked config* and treat val-selection as optional sensitivity, not a second recipe. LBP reaches LiRA 0.530 at Acc 0.621; Fig. 4 makes the Acc–LiRA tradeoff explicit.

A Cora GraphSAGE check with n_{shadows}=8 (3 seeds) preserves—and strengthens—the ordering: none LiRA 0.704 → SAMI 0.551 at Acc 0.879 → 0.873; confirmatory tables use n=4 (citation) and n=2 (ogbn smoke wall-clock), never attack-tuned.

D. Intervention validity (−LTE)

Fig. 5 / Table X: removing LTE collapses Cora Acc by ≈0.11 while conf AUROC barely moves (0.574 vs. 0.571). So −LTE validates that risk weighting is needed for *utility under alignment*, not that LTE is a calibrated privacy targeting mechanism. HCAG and release noise remain optional knobs.

E. ogbn-arxiv: systems + negative control (not a threat proof)

Table XII: NeighborLoader GraphSAGE on ∼169k nodes. Under the official OGB split, LiRA/conf sit near chance (n_{shadows}=2 in the main systems grid). Table XIX: none+SAMI at n_{shadows}=4 × 3 seeds remains LiRA ≈0.503—a credibility check that the negative control is not an artifact of underpowered shadows. This validates that the audit *scales* and that *this protocol finds little score MIA here*; it does not prove large graphs are safe in general, nor that “Volume = leakage.” Volume-safe GTD (stage-1 CE) matches undefended Acc after an unstable NeighborLoader pseudo stage collapsed Acc—we report the fair CE peer, not the collapsed artifact. LBP taxes Acc without buying LiRA.

a) Amazon Computers (multi-k bridge, not Volume).: Table XI: ∼13.7k-node co-purchase graph under the same 40/20/40 protocol. LiRA sits at chance for *all* defenses (≈0.50); Acc remains usable (≈0.71 undefended). This is a second real multi-k *negative control*—scale alone does not create score MIA—not a privacy-win cell for SAMI.

b) Controlled stress test (not a Volume peer).: Table XIII is a synthetic stress cell (n=3k, Acc≈0.29, gen-gap≈0.41): conf AUROC is high while **LiRA stays at chance**. We treat it as a controlled overfitting stress test for conf-based scores—not evidence that Volume graphs leak, and not commensurate with ogbn on the regime map. Likely conf≫LiRA causes: diffuse posteriors under failed Acc, shadow-likelihood noise, and membership signal concentrated in max-confidence features (see `lte_mechanism_audit.json`).

On Actor, Acc is low for *all* methods (≈0.29–0.34). Honest LiRA ranking under locked σ=0.35: LBP (0.543) < SAMI (0.552) < GTD (0.585) < none (0.605) ≪ MaskArmor (0.797). Pre-registered higher-σ retune (val Acc ≥ LBP_{val}+0.03, then max σ): selects $(\lambda, \sigma) = (1.0, 0.5)$ and *beats LBP on LiRA* (0.537 vs. 0.543) at higher Acc (0.306 vs. 0.289); the joint bar Acc≥LBP+0.03 is not met (ΔAcc= +0.018). We therefore report Actor as: locked SAMI improves over none/GTD/MaskArmor; val-selected SAMI can edge LBP on LiRA without matching the Acc margin bar—not a clean joint-frontier win.

F. Chameleon: second heterophilic real grid

Table XV: locked SAMI on Wikipedia Chameleon (h≈0.24). LiRA 0.623 → 0.551 (Δ=−0.071) with Acc 0.449 → 0.460 vs. none—a second real-data LiRA win at matched-or-better Acc. vs. GTD (LiRA 0.581) and MaskArmor (LiRA 0.774 inflate); LBP remains slightly lower on LiRA (0.544) at much worse Acc (0.389). Together with Cora GraphSAGE, this is the multi-regime privacy story.

G. LTE is a proxy (with quintile attribution)

Table XVI: Spearman(r_v, $|\phi_2|$–gap) is weak (Cora/Citeseer modest; Actor/Chameleon ≈0)—LTE is not a calibrated leakage estimator. Table XVII: under locked SAMI the conf-AUROC *change* redistributes—high-LTE quintiles improve

TABLE VII
MAIN RESULTS (MEAN OVER 5 SEEDS). CITESEER GRAPHSAGE SAMI = LOCKED CONFIG (MATCHES TABLE VI); GCN COLUMNS FROM CONFIRMATORY CORE.

Dataset	Defense	GraphSAGE				GCN			
		Acc ↑	Conf ↓	LiRA ↓	TPR@1% ↓	Acc ↑	Conf ↓	LiRA ↓	TPR@1% ↓
Cora	none	0.877	0.650	0.567	0.033	0.865	0.581	0.539	—
Cora	label smoothing	0.879	0.712	0.591	0.038	0.868	0.616	0.549	—
Cora	DropEdge	0.874	0.680	0.575	—	0.869	0.573	0.537	—
Cora	GTD-style	0.876	0.585	0.538	0.014	0.874	0.565	0.535	—
Cora	LBP	0.701	0.579	0.645	0.021	0.683	0.562	0.562	—
Cora	MaskArmor	0.877	0.562	0.580	0.000	—	—	—	—
Cora	**SAMI**	0.874	**0.563**	**0.510**	**0.003**	0.862	0.567	**0.533**	—
Citeseer	none	0.740	0.741	0.596	0.045	0.717	0.641	0.560	—
Citeseer	GTD-style	0.739	0.629	0.551	0.022	0.724	0.653	0.546	—
Citeseer	LBP	0.621	0.629	0.530	0.028	0.596	0.627	0.593	—
Citeseer	**SAMI**	0.731	0.642	**0.537**	0.026	0.724	0.633	0.575	—

TABLE VIII
GCN HARD CELL ON `SYNTHETIC_LOW_SPARSE` (5-SEED CONFIRM; ARCH-AWARE LTE). SUCCESS BAR: ΔCONF/LIRA ≤ -0.03 WITH ΔACC ≥ -0.03 — NOT MET.

Defense	Acc	Conf	LiRA
none	0.773	0.587	0.502
SAMI (best)	0.746	0.593	0.499

TABLE IX
LOCKED SAMI HYPERPARAMETERS FOR CORA/CITESEER/CHAMELEON/ACTOR (OGBN: σ=0.1, HCAG OFF). VAL-SELECTION ON CITESEER MATCHES LOCKED LIRA; NO TEST-ATTACK TUNING.

Knob	Value
λ, warmup, σ	0.5, 5, 0.35
LTE / arch-aware / HCAG	on / on / on (citation); HCAG off at ogbn
entropy coef, budget B	0.05, 0

TABLE X
ABLATIONS ON CORA (MEANS, 5 SEEDS).

Model	Variant	Acc	Conf
GraphSAGE	full SAMI	0.881	0.574
GraphSAGE	−LTE	0.772	0.571
GraphSAGE	none	0.877	0.650
GCN	full SAMI	0.862	0.567
GCN	−LTE	0.752	0.561

TABLE XI
AMAZON COMPUTERS GRAPHSAGE (5 SEEDS; n_{shadows}=4). NEAR-CHANCE LIRA ACROSS THE BOARD.

Defense	Acc ↑	LiRA ↓	Conf ↓	Train s
none	0.706	0.499	0.511	28
GTD-style	0.505	0.501	0.508	33
LBP	0.425	0.500	0.503	38
MaskArmor	0.706	0.502	0.506	34
SAMI (locked)	0.651	0.500	0.506	118

TABLE XII
OGBN-ARXIV GRAPHSAGE (3 SEEDS; n_{shadows}=2). SYSTEMS + PRIVACY.

Defense	Acc ↑	Conf ↓	LiRA ↓	Train s
none	0.666	0.527	0.502	96
SAMI	0.643	0.528	0.499	108
GTD-style	0.664	0.526	0.501	197
LBP	0.454	0.508	0.499	104

TABLE XIII
CONTROLLED STRESS TEST (NOT VOLUME PEER): GCN n=3K, LOW-h/LOW-SNR; ACC COLLAPSES; LIRA≈CHANCE WHILE CONF AUROC IS HIGH.

Defense	Acc ↑	Conf ↓	LiRA ↓	Gen. gap
none	0.291	0.794	0.494	0.414
SAMI	0.261	0.682	0.502	0.242

TABLE XIV
ACTOR GRAPHSAGE BASELINES (5 SEEDS; n_{shadows}=4; LOCKED SAMI). ACC IS LOW ON THIS HETEROPHILIC GRAPH FOR ALL METHODS.

Defense	Acc ↑	LiRA ↓	Conf ↓	Train s
none	0.335	0.605	0.875	3.5
GTD-style	0.339	0.585	0.847	2.9
LBP	0.289	0.543	0.737	3.0
MaskArmor	0.335	0.797	0.790	3.1
SAMI (locked)	0.317	0.552	0.774	9.0

TABLE XV
CHAMELEON GRAPHSAGE BASELINES (5 SEEDS; n_{shadows}=4; LOCKED SAMI σ=0.35).

Defense	Acc ↑	LiRA ↓	Conf ↓	Train s
none	0.449	0.623	0.880	8.1
GTD-style	0.475	0.581	0.787	8.6
LBP	0.389	0.544	0.761	8.8
MaskArmor	0.449	0.774	0.757	7.5
SAMI (locked)	0.460	0.551	0.789	25.3

(Cora/Actor Q5 $\Delta\approx-0.15$) while low-LTE Q1 can worsen (Cora $+0.12$). We frame LTE as budgeting that spends privacy effort on high-r_v nodes, not as uniform privacy for all nodes.

TABLE XVI
LTE vs. $|\phi_2|$–GAP SPEARMAN (UNDEFENDED GRAPHSAGE).

Dataset	Spearman	p
Cora	0.077	$< 10^{-3}$
Citeseer	0.118	$< 10^{-8}$
Actor	-0.007	0.57
Chameleon	-0.057	$< 10^{-2}$

TABLE XVII
CONF AUROC BY LTE RISK QUINTILE (3 SEEDS; GRAPHSAGE). Δ = SAMI$-$NONE.

Dataset	Q1 Δ	Q3 Δ	Q5 Δ	Q5$-$Q1
Cora	$+0.12$	-0.10	-0.15	-0.27
Actor	0.00	-0.07	-0.15	-0.15

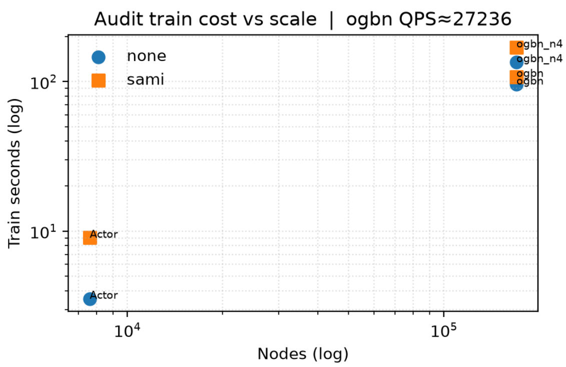

Fig. 7. Audit train cost vs. graph size (none/SAMI). Systems pillar stands even when Volume LiRA is near chance.

H. How to audit your API (SPAB)

1) Fix the membership predicate (supervised mask vs. graph visibility).
2) Measure (h, ρ) and choose M; record n and split protocol.
3) Run defense-aware confidence + LiRA (report TPR@1%FPR); log train s / RSS / QPS.
4) Emit one SPAB row: $(h, \rho, M, n, \mathrm{Acc}, \mathrm{LiRA}, \mathrm{TPR}, \mathrm{s}, \mathrm{MB}, \mathrm{QPS})$.
5) If LiRA $\gg$ 0.5 and Acc budget allows, apply risk-weighted hardening (SAMI) or DP; re-audit.

Fig. 7 summarizes NeighborLoader audit train cost vs. n (ogbn QPS$\approx$27k).

I. Calibration (secondary)

Label smoothing worsens ECE and conf AUROC on Cora GraphSAGE (Table XVIII). Label-only gap moves only marginally under SAMI—strong claims target score-based APIs. The synthetic LOO plot (Fig. 1) remains supporting evidence only.

J. Protocol limitations

- **Splits.** Citation/Actor use 40/20/40 random masks (not Planetoid public splits) for seed-wise membership resampling. On the public Planetoid Cora split (5 seeds; n_{shadows}=4), locked SAMI still improves LiRA 0.532 $\to$ 0.511 at Acc 0.788 $\to$ 0.837 (see `planetoid_cora_appendix_summary.json`).
- **ogbn LiRA n_{shadows}.** Main Volume grid uses n=2 (smoke budget); Table XIX reports none+SAMI at n=4$\times$3 seeds for credibility.
- **Metric emphasis.** Protocol primary = LiRA; some plots/stress cells show conf AUROC—we label them secondary.
- **SAMI/val.** Training uses val as the non-member proxy for AdvReg/MMD (never test); this matches defense-aware evaluation but couples the alignment objective to the split.
- **Volume GTD.** We report stage-1-only CE after NeighborLoader pseudo-label collapse; this is a fair Acc peer, not a claim that full two-stage GTD transfers unchanged.
- **ogbn-products.** Cached ($\sim$2.45M nodes). Full Acc+MIA audits OOM on a 16GB CPU host (exit 137). Systems-only NeighborLoader probe: Acc$\approx$0.69 (50k test subsample), train $\approx$182s / 5 epochs, peak RSS $\approx$7.4GB, QPS$\approx$3.3k (`ogbn_products_systems_only.json`). Amazon Computers ($\sim$13.7k) is the multi-k bridge grid. Volume privacy claims remain ogbn-arxiv.
- **Locked σ.** Paper tables use σ=0.35 (Table IX) for citation/Actor/Chameleon; ogbn SAMI uses NeighborLoader config (σ=0.1, HCAG off).

TABLE XVIII
CORA GRAPHSAGE: CALIBRATION AND PRIVACY.

Defense	Acc	ECE	Conf	LiRA
none	0.877	0.051	0.650	0.567
label smoothing	0.879	0.111	0.712	0.591
LBP	0.701	0.132	0.579	0.645
GTD-style	0.876	0.092	0.585	0.538
SAMI	0.874	0.097	0.563	0.510

TABLE XIX
OGBN-ARXIV LIRA CREDIBILITY: NONE+SAMI, n_{shadows}=4, 3 SEEDS (OFFICIAL SPLIT). STILL NEAR CHANCE.

Defense	Acc	Conf	LiRA
none	0.666	0.527	0.503
SAMI	0.643	0.528	0.503

VIII. DISCUSSION

a) Operators.: Treat privacy as workload-dependent: emit one SPAB row per (h, ρ, M, n) and spend hardening budget only where LiRA$\gg$ 0.5 and Acc allows. NeighborLoader systems numbers (train s, RSS, QPS) belong in the same row as Acc/LiRA so audit cost is auditable.

b) Open problems.: A Volume-scale graph with LiRA>0.55 remains open; under present evidence the Big Data claim is *null at scale* (ogbn n=4, Computers) plus systems cost. Learned (val-only) risk weights that beat LTE on ϕ-gap correlation remain open.

c) Artifact.: SPAB v1.0 schema (`SPAB_v1_SPEC.md`), frozen `results/paper_release/`, and `build_spab_release.py` target byte-identical CSV regeneration. Zenodo deposit accompanies the camera-ready.

IX. Conclusion

SPAB is a scale-aware conditional audit protocol: ogbn-arxiv contributes NeighborLoader systems and a trustworthy near-chance negative control under this split—not a demonstration that large graphs always leak. Privacy findings are concentrated on locked-config SAMI for Cora/Citeseer/Chameleon (Fig. 4); Actor shows when LBP wins LiRA alone; Volume negatives (ogbn, Computers) show scale≠leakage under this protocol. Artifacts: `spab_report.csv`, protocol limitations, and frozen release files.

Code Availability

Configuration, defenses, SCML/Volume scripts, and frozen artifacts: https://github.com/PrivacyAI-Lab/privacygnn. See `REPRODUCE.md` and `EVALUATION_PROTOCOL.md`. Zenodo deposit package: `results/spab_v1_zenodo/` (run `bash build_zenodo_package.sh`; set DOI via `scripts/set_zenodo_doi.py` after publish).

Reproducibility Checklist

- **SPAB report:** `results/spab_report.csv` / `.json` (matches paper tables).
- **Config:** `experiment_config_confirmatory.yaml`; Volume: `experiment_config_ogbn.yaml`.
- **Protocol:** `EVALUATION_PROTOCOL.md`.
- **Commands:** `run_core_tables.py`; `run_highrisk_volume_synth.py`; `build_spab_release.py`; `freeze_paper_release.py`.
- **Hardware:** CPU confirmatory runs; LiRA shadow counts protocol-sized by wall-clock.

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